

1. Prerequisites

Ensure you have:

- **Java 8+** installed.
 - **AMPS** installed and running.
 - **OpenSSL** (optional, for verification).
 - **Keytool** (bundled with Java).
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2. Generating Certificates Using Java Keytool

We will create:

- **A self-signed CA certificate.**
- **A server certificate (signed by CA)** for AMPS.
- **A client certificate (signed by CA)** for mutual authentication.

Step 1: Generate a Self-Signed CA Certificate

```
keytool -genkeypair -alias amps-ca \  
-keyalg RSA -keysize 2048 -validity 365 \  
-keystore amps-ca.jks \  
-dname "CN=AMPS-CA, OU=IT, O=MyCompany, L=City, ST=State, C=US" \  
-storepass capass -keypass capass
```

This creates a **Certificate Authority (CA)**.

Step 2: Export the CA Certificate

```
keytool -export -alias amps-ca \  
    -keystore amps-ca.jks \  
    -file amps-ca.crt \  
    -storepass capass
```

This file **amps-ca.crt** will be used to sign the server and client certificates.

3. Create & Sign Server Certificate

Step 3: Generate a KeyPair for AMPS Server

```
keytool -genkeypair -alias amps-server \  
    -keyalg RSA -keysize 2048 -validity 365 \  
    -keystore amps-server.jks \  
    -dname "CN=amps.example.com, OU=IT, O=MyCompany, L=City, ST=State, C=US" \  
    -storepass serverpass -keypass serverpass
```

Step 4: Create a Certificate Signing Request (CSR)

```
keytool -certreq -alias amps-server \  
    -keystore amps-server.jks \  
    -file amps-server.csr \  
    -storepass serverpass
```

Step 5: Sign the Server Certificate with CA

```
keytool -gencert -alias amps-ca \  
-keystore amps-ca.jks \  
-infile amps-server.csr \  
-outfile amps-server.crt \  
-storepass capass
```

Step 6: Import the CA Certificate into the Server Keystore

```
keytool -import -trustcacerts -alias amps-ca \  
-file amps-ca.crt \  
-keystore amps-server.jks \  
-storepass serverpass -noprompt
```

Step 7: Import Signed Certificate into the Server Keystore

```
keytool -import -trustcacerts -alias amps-server \  
-file amps-server.crt \  
-keystore amps-server.jks \  
-storepass serverpass -noprompt
```

4. Create & Sign Client Certificate (For Mutual Authentication)

Step 8: Generate Client KeyPair

```
keytool -genkeypair -alias amps-client \  
-keyalg RSA -keysize 2048 -validity 365 \  
-keystore amps-client.jks \  
-dname "CN=client.example.com, OU=IT, O=MyCompany, L=City, ST=State, C=US" \  
-storepass clientpass -keypass clientpass
```

Step 9: Create a Certificate Signing Request (CSR)

```
keytool -certreq -alias amps-client \  
-keystore amps-client.jks \  
-file amps-client.csr \  
-storepass clientpass
```

Step 10: Sign the Client Certificate with CA

```
keytool -gencert -alias amps-ca \  
-keystore amps-ca.jks \  
-infile amps-client.csr \  
-outfile amps-client.crt \  
-storepass capass
```

Step 11: Import CA Certificate into Client Keystore

```
keytool -import -trustcacerts -alias amps-ca \  
-file amps-ca.crt \  
-keystore amps-client.jks \  
-storepass clientpass -noprompt
```

Step 12: Import Signed Client Certificate into Client Keystore

```
keytool -import -trustcacerts -alias amps-client \  
-file amps-client.crt \  
-keystore amps-client.jks \  
-storepass clientpass -noprompt
```

5. Configure the AMPS Server

Edit the **AMPS** configuration file (**amps-config.xml**).

A. One-Way SSL (Server Authentication Only)

```
<AMPS>
  <Transport>
    <Name>ssl</Name>
    <Type>tcp</Type>
    <Listen>9000</Listen>
    <TLS>
      <Certificate>/path/to/amps-server.jks</Certificate>
      <KeyStorePassword>serverpass</KeyStorePassword>
      <TrustedCertificates>/path/to/amps-ca.jks</TrustedCertificates>
      <RequireClientCert>false</RequireClientCert>
    </TLS>
  </Transport>
</AMPS>
```

B. Two-Way SSL (Mutual Authentication)

```
<AMPS>
  <Transport>
    <Name>ssl</Name>
    <Type>tcp</Type>
    <Listen>9000</Listen>
    <TLS>
      <Certificate>/path/to/amps-server.jks</Certificate>
      <KeyStorePassword>serverpass</KeyStorePassword>
      <TrustedCertificates>/path/to/amps-ca.jks</TrustedCertificates>
      <RequireClientCert>true</RequireClientCert>
    </TLS>
  </Transport>
</AMPS>
```

Restart AMPS:

```
./ampServer /path/to/amps-config.xml
```

6. Configure the Java Client for SSL/TLS

A. One-Way SSL Client

```
import com.crankuptheamps.client.*;

public class SSLClient {
    public static void main(String[] args) throws Exception {
        HAClient client = new HAClient("SSLClient");

        System.setProperty("javax.net.ssl.trustStore", "/path/to/amps-ca.jks");
        System.setProperty("javax.net.ssl.trustStorePassword", "capass");

        client.connect("ssl://hostname:9000");
        System.out.println("Connected securely to AMPS");
    }
}
```

B. Mutual Authentication Client

```
import com.crankuptheamps.client.*;

public class SSLClient {
    public static void main(String[] args) throws Exception {
        HAClient client = new HAClient("SSLClient");

        System.setProperty("javax.net.ssl.keyStore", "/path/to/amps-client.jks");
        System.setProperty("javax.net.ssl.keyStorePassword", "clientpass");

        System.setProperty("javax.net.ssl.trustStore", "/path/to/amps-ca.jks");
        System.setProperty("javax.net.ssl.trustStorePassword", "capass");

        client.connect("ssl://hostname:9000");
        System.out.println("Mutually authenticated and connected to AMPS");
    }
}
```

Compile and Run:

```
javac SSLClient.java
java SSLClient
```

7. Testing

1. **Start AMPS** with SSL enabled.
 2. **Run the Java client** and verify connection.
 3. If using **Mutual Authentication**, verify client authentication is enforced.
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